

Table 1: Average per-instance inference time at $K = 53$ labels (enron), in milliseconds. The four ILP rows are the BOPOs search times plotted in the runtime figures; the lower block lists the baseline per-instance inference costs (independent of preference order and target metric). Measured GLPK entries are averaged over 1–10 test instances timed directly at $K = 53$; HiGHS and baseline values are measured over the full test set. For the pre-order search GLPK is impractically slow at full transitivity: a single $K = 53$ instance did not finish within a 48-hour wall-clock limit (vs. under a second for HiGHS), so those two entries are left † n/a. The height-2 restriction is tractable but still about three orders of magnitude slower than HiGHS. This is precisely why the full-size enron experiments use HiGHS.

Method	pre-order		partial order	
	Hamming	Subset	Hamming	Subset
GLPK, full transitivity	n/a †	n/a †	525	521
GLPK, height=2	622,640	621,091	361	351
HiGHS, full transitivity	2427	1587	1102	947
HiGHS, height=2	579	575	330	331
Binary Relevance (BR)	2.5	2.5	2.5	2.5
Classifier Chain (CC)	5.2	5.2	5.2	5.2
Calibrated Label Ranking (CLR)	30.0	30.0	30.0	30.0
Ensemble of Classifier Chains (ECC)	137	137	137	137